Skills Summary

Solving Systems by Elimination with Scaling

Use this reference while completing your worksheet or when studying.

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1. Scaling one equation

The move: multiplying *every term on both sides* of an equation by a nonzero constant gives an equivalent equation — same solutions, new coefficients. That is what makes elimination legal.

When one equation is enough: one coefficient of the target variable divides the other. Scale the smaller one up to match, then add or subtract so the target drops out.

Add or subtract? Opposite signs $\Rightarrow$ add. Same sign $\Rightarrow$ subtract.

Example: Solve $x + 2y = 11$ and $3x - y = 5$.
Step 1. The y coefficients are 2 and -1, and 1 divides 2, so scaling the second equation alone will match them — no need to touch the first.
Step 2. $2(3x - y) = 2(5) \Rightarrow 6x - 2y = 10$. Add to $x + 2y = 11$: $7x = 21$, so $x = 3$.
Step 3. Back-substitute: $3+2y = 11 \Rightarrow y = 4$. Check in $3x-y$: $9-4 = 5$ ✓ $\boldsymbol{(x, y) = (3, 4)}$

Try it: Solve $2x + y = 7$ and $x + 3y = 11$.

check: $\boldsymbol{(x, y) = (2, 3)}$

2. Scaling both equations

When neither coefficient divides the other, scale both. Take the least common multiple of the two coefficients of the target variable, and multiply each equation by what it needs to reach it.

Choose the cheaper variable first: compare the two LCMs and eliminate the variable with the smaller one. Smaller multipliers mean smaller numbers and fewer arithmetic slips.

Example: Solve $2x + 3y = 12$ and $5x + 4y = 23$.
Step 1. Neither x pair nor y pair divides evenly, so scale both; for x the LCM of 2 and 5 is 10, cheaper than the LCM of 3 and 4.
Step 2. $5(2x + 3y) = 5(12) \Rightarrow 10x + 15y = 60$ and $2(5x + 4y) = 2(23) \Rightarrow 10x + 8y = 46$. Subtract: $7y = 14$, so $y = 2$.
Step 3. Back-substitute: $2x + 6 = 12 \Rightarrow x = 3$. Check in $5x + 4y$: $15 + 8 = 23$ ✓ $\boldsymbol{(x, y) = (3, 2)}$

Try it: Solve $3x + 2y = 16$ and $4x + 5y = 26$.

check: $\boldsymbol{(x, y) = (4, 2)}$

3. Building the system from a situation

Two unknowns need two equations. Name each unknown with a letter and a unit before writing anything else.

Where the two equations come from: usually one counts things (items, litres, hours) and one measures value (dollars, salt, distance).

Then eliminate as usual, and finish by translating the ordered pair back into the words of the problem.

Example: Two adult and three child tickets cost \$37; one adult and four child tickets cost \$31. Find each price in dollars.

Step 1. Two unknown prices, two purchase facts, so name a and c in dollars and write one equation per purchase: $2a + 3c = 37$ and $a + 4c = 31$.

Step 2. The a coefficients are 2 and 1, so scale the second by 2: $2a + 8c = 62$. Subtract the first: $5c = 25$, so $c = 5$.

Step 3. Back-substitute: $a + 20 = 31 \Rightarrow a = 11$. Check: $2(11) + 3(5) = 37$ ✓ $\Rightarrow$ $\boldsymbol{(a, c) = (11, 5)}$

Try it: Three notebooks and two pens cost \$23; one notebook and four pens cost \$21. Find the price of a notebook, x, and of a pen, y, in dollars.

check: $\boldsymbol{(x, y) = (5, 4)}$

(!) Watch out: Scaling means multiplying *both sides.* Turning $x+4y = 14$ into $3x+12y = 14$ changes the equation and loses the solution; the right side must become 42.